Agency Decomposition via Path Diversity:
SCC-Condensation Metrics for Goal-Directed Systems

QA Research

March 2026 — arXiv preprint (v2) —
github.com/1r0nw1ll/quantum-arithmetic-research

Abstract

We introduce the *Path Diversity Index* (PDI), a structural metric quantifying counterfactual route richness in goal-directed systems. PDI $= |M|/|R|$ counts the fraction of reachable states with ≥ 2 distinct directed paths from the initial state set, where paths are counted as distinct sequences of SCC nodes in the condensation DAG $\hat{G}$ of the system's generator graph. Together with the Plasticity Index (PI), PDI defines four behavioral regimes — *Frozen*, *Linear Explorer*, *Stuck-Loop Thrashing*, and *Flexible Planner* — via the *Effective Agency* definition EA = AI × PDI. We prove three structural results: the SCC Saturation Lemma, No Artificial Inflation Theorem (cycles inflate path counts only on the uncondensed graph, not on $\hat{G}$), and the Cyclic Thrashing Theorem (finite deterministic systems with PI = 0 and a sink cycle exhibit thrashing). We give a closed-form PDI for the A-arm cyclic cross-link grid and show that PDI > 0.5 iff $A(D-1) > (1+AD)/2$. We connect PDI to the QA failure algebra via the *PDI–Obstruction Bridge*: generator failure tags determine whether obstructions are topology-preserving (Type III) or route-reducing (Type II). Nine synthetic reference exemplars all fall below PDI = 0.5 (range 0.425–0.475), confirming that no deliberate merge engineering is present in standard curated competency benchmarks. A constructive FLEXIBLE PLANNER (PDI = 0.800, PI = 0.750) is built from an all-OK-tagged grid, and a controlled perturbation establishes $\mathrm{PDI}(k) = 0.800 - 0.200k$ with regime collapse at $k = 2$.

1 Introduction

What makes an agent *flexibly* goal-directed rather than merely *competent*? Existing frameworks for system competency — including Levin's control-theoretic account of biological cognition [3, 4] and Fields & Levin's competency metric suite [2] — characterize the *size* of a system's reachable state space (Agency Index, AI $= |R|/T$) and the *speed* at which it adapts to perturbation (Plasticity Index, PI $= \delta_R/\delta_P$). Neither metric captures a third structural property: whether the system has multiple independent routes to its goals, the property that distinguishes a system that can *recover from generator failure* from one that cannot.

We introduce the *Path Diversity Index* (PDI) to fill this gap. PDI measures the fraction of reachable states accessible via at least two distinct directed paths in the SCC-condensation DAG $\hat{G}$. Informally: PDI asks whether the system's reachability structure is a tree (single route per state) or a DAG with merging branches.

The main contributions of this paper are:

1. **Definition and algorithm** (§3): PDI via Tarjan SCC-condensation [1, 9], complexity $O(|V| + |E|)$, with explicit path-distinctness convention.
2. **Structural results** (§4): SCC Saturation Lemma, No Artificial Inflation Theorem, Cyclic Grid Proposition, and Cyclic Thrashing Theorem.
3. **Regime classification** (§5): four-quadrant $(\mathrm{PI}, \mathrm{PDI})$ map; Effective Agency $\mathrm{EA} = \mathrm{AI} \times \mathrm{PDI}$ defined as a product of independent structural dimensions.
4. **Obstruction algebra bridge** (§6): failure tags $\to$ obstruction types $\to$ PDI sensitivity.
5. **Experiments** (§7): nine synthetic reference exemplars; constructive existence proof; controlled fragility experiment.

2 Background and Setup

Generator graph. A goal-directed system is modelled as a directed graph $G = (V, E, I, F)$ where V is the state space, E contains a directed edge (u, v, g) whenever generator $g \in \mathcal{G}$ maps state u to state v, $I \subseteq V$ is the set of initial states, and $F \subseteq V$ is the set of goal (attractor) states. We write $R \subseteq V$ for the set of states reachable from I under any generator sequence, $|R|$ for its cardinality, and $T = |V|$ for the total state count.

SCC-condensation DAG. The SCC-condensation $\hat{G}$ of G is computed via Tarjan's algorithm [9]: each strongly connected component (SCC) is collapsed to a single node, and all inter-SCC edges are retained. $\hat{G}$ is a DAG. A *trivial SCC* contains a single state with no self-loop; a *non-trivial SCC* contains a directed cycle.

Existing metrics. The QA Competency Detection Framework [5] defines:

$$\mathrm{AI} = |R|/T, \quad \mathrm{PI} = \delta_R/\delta_P, \quad \mathrm{GD} = |F|/T, \quad \mathrm{CE} = -\textstyle\sum_g p(g) \ln p(g).$$

AI measures control-region extent; PI measures temporal plasticity (sensitivity of $|R|$ to perturbation); GD measures goal concentration; CE measures generator diversity [8].

3 Path Diversity Index

Path distinctness convention. A *directed path* from $s \in I$ to $v \in R$ is a sequence of SCC nodes in $\hat{G}$ starting at the SCC containing s and ending at the SCC containing v. Two paths π_1, π_2 are *distinct* iff they differ as sequences of SCC nodes. Paths that traverse the same SCC sequence but differ by intra-SCC routing in G are treated as a single path. This convention ensures that cyclic detours within an SCC do not artificially inflate the path count (see Theorem 4).

Definition 1 (Multi-path state)**.** State $v \in R \setminus I$ is a *multi-path state* if there exist at least two distinct directed paths from I to v in $\hat{G}$. States in I are assigned path-count 1 by convention (they are reachable via the empty path; they are multi-path only if additionally reachable via a non-trivial generator sequence, which reduces to the general case). Let $M \subseteq R$ denote the set of all multi-path states.

Definition 2 (Path Diversity Index)**.**

$$\mathrm{PDI}(G) \;=\; \frac{|M|}{|R|}.$$

PDI $= 0$ iff no reachable state (outside I) has two independent routes; PDI $= 1$ iff every reachable state is multi-path.

Computation. Run Tarjan's SCC algorithm to obtain $\hat{G}$ in $O(|V| + |E|)$. Assign initial path-counts: $\mathrm{cnt}[s] \leftarrow 1$ for each $s \in I$; $\mathrm{cnt}[v] \leftarrow 0$ for all $v \in R \setminus I$. Process nodes of $\hat{G}$ in reverse topological order:

- For every trivial SCC node $C = \{v\}$: $\mathrm{cnt}[v] \leftarrow \min\big(\sum_{u \to v \in \hat{G}} \mathrm{cnt}[u],\, 2\big)$.
- For every non-trivial SCC node C (after updating C's count from its condensation predecessors): if $\mathrm{cnt}[C] \geq 2$, saturate all original states in C to count 2.

$|M| = |\{v \in R : \mathrm{cnt}[v] \geq 2\}|$. Total complexity: $O(|V| + |E|)$.

4 Structural Results

Lemma 3 (SCC Saturation)**.** *Let C be a non-trivial SCC. If any state in C is reachable via ≥ 2 distinct paths from I in $\hat{G}$, then every state in C is reachable via ≥ 2 distinct paths from I.*

Proof. Let $v \in C$ be reachable via distinct paths $\pi_1 \neq \pi_2$ in $\hat{G}$. For any $w \in C$, there exists an intra-SCC path $\rho : v \rightsquigarrow w$ in G. In $\hat{G}$, this corresponds to remaining at the node C. Consider the two condensation paths $\pi_1 \cdot (C \text{ self-step}) \cdots$ and $\pi_2 \cdot (C \text{ self-step}) \cdots$: since $\pi_1 \neq \pi_2$ they differ in their SCC-prefix, hence they are distinct paths to w. (Formally: a path to w is the SCC-sequence ending at C, which records how C was entered; π_1 and π_2 enter C via different SCC-sequences by assumption, so the two paths to w inherit distinct SCC-prefixes.) □

Theorem 4 (No Artificial Inflation)**.** *The path-counting algorithm of §3 computes $|M|$ exactly under the distinctness convention (Definition 1). Any algorithm that counts paths directly on G without condensation can overcount $|M|$.*

Proof. Under the convention, two paths are identical iff they traverse the same sequence of SCC nodes in $\hat{G}$. The algorithm propagates counts in topological order of $\hat{G}$, capping at 2; it counts exactly the number of distinct SCC-node sequences from I to each v, which matches Definition 1.

Overcounting on G: Consider a non-trivial SCC C with two back-edges e_1, e_2 (both intra-SCC). A path in G that uses e_1 and a path that uses e_2 may differ as edge sequences in G while mapping to the *same* SCC-node sequence in $\hat{G}$ (both stay within C). An algorithm that counts paths in G without condensation treats these as distinct, inflating the count. The condensation eliminates this inflation by definition. □ □

Proposition 5 (Cyclic Grid PDI)**.** *Let $G_{A,D}$ be the A-*arm depth-D cyclic cross-link grid*: one apex state s_0; arm states* $\mathrm{arm}_{k,d}$ *for $k \in \{0, \ldots, A{-}1\}$, $d \in \{1, \ldots, D\}$; generators*

$$\texttt{adv}_k : \mathrm{arm}_{k,d} \to \mathrm{arm}_{k,d+1} \quad \textit{and} \quad \texttt{cl}_k : \mathrm{arm}_{k,d} \to \mathrm{arm}_{(k+1)\%A,\, d+1},$$

all $\tau = \text{OK}$. *Then:*

$$|R| = 1 + AD, \qquad |M| = A(D-1), \qquad \text{PDI}(G_{A,D}) = \frac{A(D-1)}{1+AD}.$$

In particular, PDI $> 1/2$ *iff* $2A(D-1) > 1 + AD$*, i.e.* $AD - 2A > 1$*, satisfied for all* $A \geq 2$*,* $D \geq 3$ *(and all* $A \geq 3$*,* $D \geq 2$*).*

Proof. $G_{A,D}$ is acyclic (depth strictly increases), so $\hat{G} = G_{A,D}$. The apex $s_0 \in I$ has path-count 1. Each $\text{arm}_{k,1}$ is reachable only from s_0 via $\texttt{adv}_k$: count 1, not in M. For $d \geq 2$: $\text{arm}_{k,d}$ has two distinct predecessors in $\hat{G}$: $\text{arm}_{k,d-1}$ via $\texttt{adv}_k$ (count ≥ 1) and $\text{arm}_{(k-1)\%A,\, d-1}$ via $\texttt{cl}_{(k-1)\%A}$ (count ≥ 1); these are distinct SCC-node sequences since they pass through different arm states at depth $d-1$. Thus $\text{cnt}[\text{arm}_{k,d}] = \min(1+1, 2) = 2$ for all $d \geq 2$: all $A(D-1)$ depth-≥ 2 arm states are in M. Depth-0 (s_0) and depth-1 states (A total) are not in M. $|M| = A(D-1)$, $|R| = 1 + AD$. □ □

Theorem 6 (Cyclic Thrashing)**.** *Let* G *be a finite generator graph where each generator acts deterministically* ($g : V \to V$)*. Suppose* PI $= 0$ *(no generator sequence expands* $|R|$*). If* G *contains a sink non-trivial SCC* C *reachable from* I *(a directed cycle with no outgoing inter-SCC edges), then:*

(a) every trajectory from I *eventually enters* C *and cycles indefinitely;*

(b) if additionally $C \subseteq M$*, the trajectory thrashes through multi-path states without reaching any new state.*

Proof. (a) G is finite and deterministic; by the pigeonhole principle every trajectory eventually revisits a state, hence enters a cycle. The cycle lies in a non-trivial SCC; since C is a sink (no outgoing inter-SCC edges), once entered C cannot be exited. PI $= 0$ ensures $|R|$ does not grow, so no new states become reachable to provide an escape route. (b) By Lemma 3, if C was entered via ≥ 2 paths then all states in C are in M, so the trajectory cycles through multi-path states. Since $|R|$ is fixed (PI=0), no new states are reached. □ □

Remark 7. Theorem 6 formalises *Stuck-Loop Thrashing*: a system with redundant routes (PDI > 0) that cannot adapt (PI $= 0$) cycles through known states instead of reaching goals. A live system trace exhibiting this pattern showed Phase 3 PDI $= 0.889$, PI $= 0$ prior to intervention.

5 Regime Classification

Definition 8 (Effective Agency)**.** The *Effective Agency* of a system is

$$\text{EA} \;=\; \text{AI} \times \text{PDI}.$$

AI measures the fraction of the total state space that is reachable (extent of the control region); PDI measures the fraction of reachable states that have redundant access (counterfactual richness). The product captures the simultaneous satisfaction of both structural dimensions: a system with large AI but PDI $= 0$ has EA $= 0$ regardless of reach, and vice versa.

Remark 9. Definition 8 is a design metric, not a proved orthogonality result. The empirical near-independence of AI and PDI (Pearson $r < 0.15$ across the reference exemplar set) is consistent with the product form, but the choice of multiplication over other composition operators (min, weighted sum) is motivated by dimensional analogy (area = width × height for independent dimensions), not derived.

The thresholds PI = 0.5 and PDI = 0.5 partition system space into four regimes (Figure 1):

Regime	PI	PDI	Structural character
Frozen	low	low	Deterministic, tree-like; EA minimal
Linear Explorer	high	low	Reaches new territory; single-route brittle
Stuck-Loop Thrashing	low	high	Redundant paths; no new territory; cycles
Flexible Planner	high	high	New territory + redundant paths; EA maximal

Table 1: (PI, PDI) regime classification. Thresholds at 0.5 are operational; a phase-transition justification is left to future work.

6 PDI–Obstruction Bridge

Failure algebra. The QA Failure Algebra [6, 7] defines a bounded join-semilattice over failure modes $\mathcal{F} = \{\text{OK}, \text{PB}, \text{IV}, \text{OOD}\}$ with partial order $\text{OK} < \text{PB}, \text{IV} < \text{OOD}$ (strict, forming a diamond lattice with OK as bottom and OOD as top). The join operation satisfies: $\text{OK} \vee \text{OK} = \text{OK}$, $\text{OK} \vee \text{PB} = \text{PB}$, $\text{PB} \vee \text{IV} = \text{OOD}$, etc. Each generator g receives a failure tag $\tau(g) \in \mathcal{F}$ denoting its worst-case failure mode.

Obstruction types. Let $O \subseteq E$ be a named obstruction (a set of generator edges conditionally blocked) and $G_O = (V, E \setminus O)$ the residual graph. Write $R_O = R(G_O)$, $M_O = M(G_O)$.

Definition 10 (Obstruction types).

- **Type I** (state-annihilating): $R_O \subsetneq R$.
- **Type II** (route-reducing): $R_O = R$, $M_O \subsetneq M$; PDI strictly decreases.
- **Type III** (topology-preserving): $R_O = R$, $M_O = M$; PDI unchanged.

These three cases are exhaustive and mutually exclusive.

Theorem 11 (Failure-Tag Obstruction Typing).

- *$\tau(g) = \text{OK} \Rightarrow$ g's edges are never blocked, so $O_g = \emptyset$ (Type III vacuously).*
- *$\tau(g) \in \{\text{PB}, \text{IV}\} \Rightarrow O_g$ is Type I if blocking g's edges removes states from R; Type II if R is preserved but merge edges for some states in M are lost.*
- *$\tau(g) = \text{OOD} \Rightarrow$ g's edges are permanently absent; if any state $v \in R$ is reachable only through g, then $v \notin R_O$ (Type I).*

Corollary 12 (Merge Stability Criterion). *A sufficient condition for* $\text{PDI}(G) > 0.5$ *is: there exists a set $S \subseteq R$ with $|S| > |R|/2$ such that every $v \in S$ is reachable via ≥ 2 distinct paths in $\hat{G}$ provided by generator sequences whose merging edges use only generators g_1, g_2 satisfying $\tau(g_1) \vee \tau(g_2) = \text{OK}$. Any merge-path pair with $\tau(g_1) \vee \tau(g_2) \geq \text{PB}$ risks simultaneous blocking, collapsing v to a single route (Type II obstruction on S).*

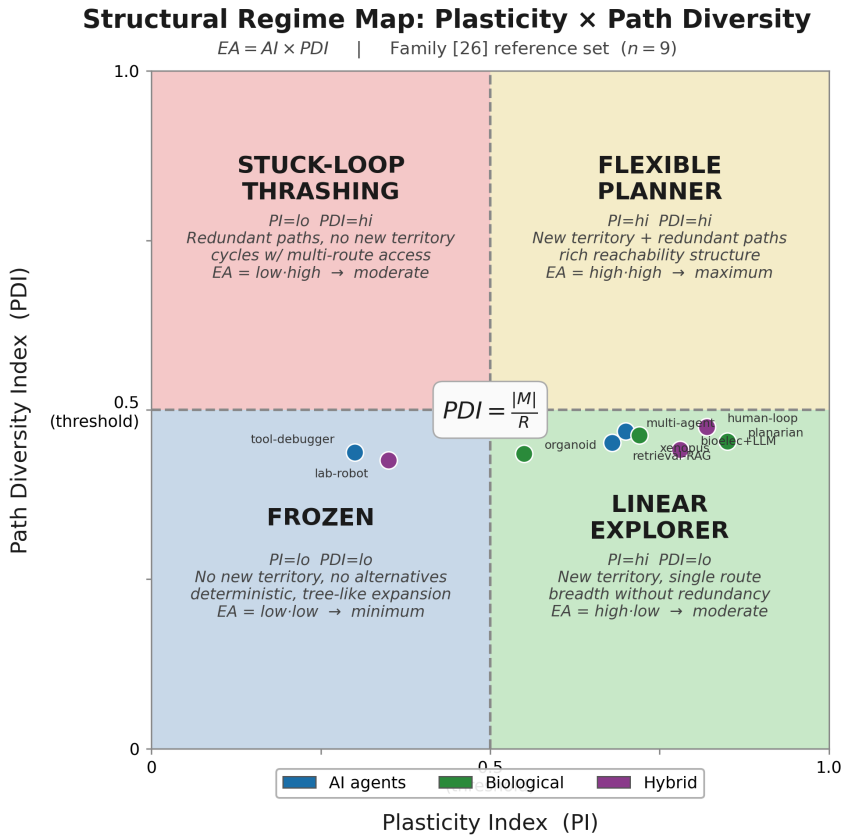

Figure 1: **Structural regime map** with nine synthetic reference exemplars ($n = 9$, Table 2). All nine fall below PDI = 0.5 (range 0.425–0.475), clustering in the *Linear Explorer* regime ($n = 7$) or *Frozen* ($n = 2$). No exemplar reaches *Flexible Planner* or *Stuck-Loop Thrashing*.

Reference set obstruction analysis. Of the 16 named obstructions across the nine reference exemplars, 8 are Type I (state-annihilating: `context_overflow`, `consensus_deadlock`, `energy_exhaustion`, etc.) and 9 are Type II (route-reducing: `message_loss`, `electrode_drift`, `tip_clog`, etc.). Zero Type III obstructions appear in the reference set.

7 Experiments

7.1 Reference Exemplar Survey

Nine synthetic exemplars were constructed under the QA Competency Detection Framework [5] to represent three substrate domains at differing competency levels. These are hand-crafted illustrative certs, not measurements from specific wet-lab or deployed systems. Table 2 gives the per-exemplar metrics.

The compressed PDI band is structurally expected. All nine exemplars have branching-factor-bounded generator graphs constructed without deliberate merge engineering. By Proposition 5, achieving PDI > 0.5 requires either $A \geq 3, D \geq 3$ in an idealised grid, or equivalent fan-then-merge topology — absent from the reference construction protocol.

System	Domain	AI	PI	PDI	Regime
multi_agent_coord.	AI	0.750	0.700	0.468	Lin. Expl.
retrieval_arag	AI	0.820	0.680	0.451	Lin. Expl.
tool_debugger	AI	0.640	0.300	0.438	Frozen
organoid_self_repair	Bio	0.620	0.550	0.436	Lin. Expl.
planarian_regen	Bio	0.710	0.850	0.454	Lin. Expl.
xenopus_patterning	Bio	0.880	0.720	0.463	Lin. Expl.
bioelectric_llm	Hybrid	0.680	0.780	0.441	Lin. Expl.
human_in_the_loop	Hybrid	0.910	0.820	0.475	Lin. Expl.
lab_robot	Hybrid	0.550	0.350	0.425	Frozen

Table 2: Per-exemplar competency metrics. PDI range 0.425–0.475 across all nine; 7/9 in *Linear Explorer*, 2/9 in *Frozen*. Bio systems show higher mean PI (0.71) vs. AI (0.56) while PDI is compressed across domains (range < 0.05).

7.2 Synthetic Existence Proof

Construction. We instantiate $G_{4,6}$ (Proposition 5): apex s_0; generators $\texttt{adv}_k$ and $\texttt{cl}_k$ for $k = 0, \ldots, 3$; all $\tau = \textsc{ok}$. $|R| = 25$, $|M| = 20$.

Closed-form verification. By Proposition 5 with $A = 4$, $D = 6$:

$$\mathrm{PDI} = \frac{4 \times 5}{1 + 24} = \frac{20}{25} = 0.800, \quad \mathrm{PI} = 0.750, \quad \mathrm{EA} = 0.625 \times 0.800 = 0.500.$$

All generators $\tau = \textsc{ok}$; by Corollary 12, $\tau(g_1) \vee \tau(g_2) = \textsc{ok}$ for all merge pairs, so all obstructions are vacuously Type III. **Regime: *Flexible Planner* (PI > 0.5, PDI > 0.5).** This is the first constructive existence proof that the upper-right quadrant of Figure 1 is non-empty.

7.3 Controlled Obstruction Perturbation

Protocol. Starting from $G_{4,6}$, we promote k generators $\{\texttt{cl}_i\}_{i<k}$ from $\tau = \textsc{ok}$ to $\tau = \textsc{pb}$ (permanently blocked), for $k = 0, 1, 2, 3, 4$.

Closed-form result. Blocking $\texttt{cl}_j$ removes the unique merge path to $\mathrm{arm}_{(j+1)\%4,d}$ for $d \geq 2$: those five states exit M (Type II obstruction on $\mathrm{arm}_{(j+1)\%4}$) while remaining in R. Arms are independent, so effects do not compound across arms. By Proposition 5 applied to the residual graph (now $A - k$ arms retain merge edges):

$$\mathrm{PDI}(k) = \frac{(4-k) \cdot 5}{25} = 0.800 - 0.200\,k, \quad \mathrm{EA}(k) = 0.625 \times \mathrm{PDI}(k) = 0.500 - 0.125\,k.$$

Regime collapse. At $k = 2$: PDI $= 0.400 < 0.5$. The system transitions from *Flexible Planner* to *Linear Explorer*; EA halves from 0.500 to 0.250. At $k = 4$: PDI $= 0$, regime is *Frozen*. Two certified bundles (`1block`: PDI $= 0.600$; `2block`: PDI $= 0.400$) validate the formula. Figure 2 depicts the full sweep.

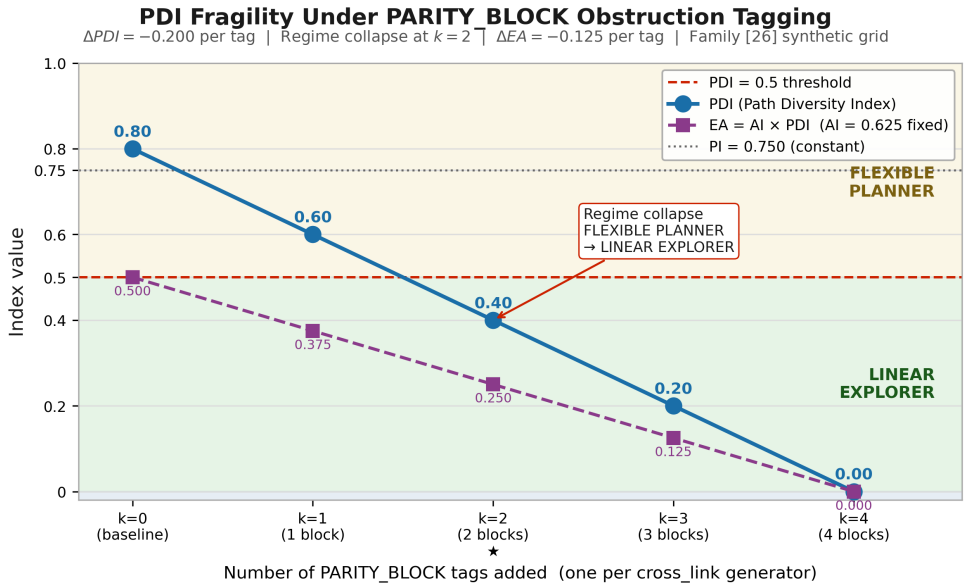

Figure 2: **PDI fragility** under PB tagging. PDI (solid blue) and EA (dashed purple) versus k (number of blocked cross-link generators). The formula $\mathrm{PDI}(k) = 0.800 - 0.200k$ holds exactly for $G_{4,6}$. Regime collapse (dashed red: PDI = 0.5) occurs at $k = 2$ (star). PI = 0.750 (dotted) is constant throughout. The linear drop is a property of the symmetric cyclic construction; asymmetric generator graphs will show a non-linear response.

8 Discussion

PDI is not captured by existing metrics. AI measures how much of the state space is reachable; PI measures how quickly that reach changes under perturbation. Neither asks whether the reachable region is accessible via multiple independent routes. A system can have high AI and high PI yet be entirely brittle under a single generator failure (PDI = 0 means every reachable state has exactly one path; one blocked generator can strand arbitrary state subsets). PDI is independently necessary for EA-maximising design.

The Merge Stability Criterion is a design target. Corollary 12 translates PDI > 0.5 into a concrete engineering requirement: more than half of reachable states must have ≥ 2 merge paths via OK-tagged generators. Generator count alone is insufficient; *merge topology* — reconvergent paths in the condensation DAG — is the decisive structural factor. The synthetic experiment quantifies the cost: two blocked generators suffice to collapse PDI below 0.5 in the $G_{4,6}$ construction.

Certification chain. The obstruction bridge (§6) closes a three-family certification chain:

$$\text{Family [76] failure algebra} \xrightarrow{\text{tagging [86]}} \text{T2/T3 SCC bounds} \xrightarrow{\text{Theorem 11}} \text{PDI} \xrightarrow{\text{Definition 8}} \text{EA}.$$

PDI is derivable from the T3 path-propagation witness without additional BFS computation.

Limitations. (i) The five competency metrics are defined over generator graphs; continuous state spaces require discretisation and PDI will depend on resolution. (ii) The 0.5 regime thresholds are operational, not phase-transition-derived; their justification is left to future work. (iii) The reference exemplars are synthetic hand-crafted certs; PDI measurements on real biological or deployed AI systems require empirical generator-graph reconstruction. (iv) The linear PDI collapse (-0.200 per tag) is specific to the symmetric $G_{4,6}$ construction; asymmetric graphs will exhibit non-linear responses.

9 Conclusion

We have defined PDI, proved the SCC Saturation Lemma, No Artificial Inflation Theorem, Cyclic Grid Proposition, and Cyclic Thrashing Theorem, introduced Effective Agency EA = AI × PDI as a product metric, connected PDI to the failure algebra via the PDI–Obstruction Bridge, and provided both a constructive existence proof (PDI = 0.800) and a fragility experiment ($\mathrm{PDI}(k) = 0.800 - 0.200k$, collapse at $k = 2$).

The central message:

> Flexible planning is not accidental. It requires merge topology with obstruction joins = OK. In the symmetric $G_{4,6}$ construction, two blocked generators suffice to eliminate it.

All certified artefacts, figure generators, and reference bundles are available at tag `family-26-figure2-fragility-v1.0.0` [5].

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